NLP COURSE PROJECT · IIIT GUWAHATI

LLM-as-a-Judge

An evaluation pipeline for helpfulness, correctness and tone, validated against human ratings

Tushar 2301234 Priyanshu kumar 2301163 Shashank kumar 2301195

Grading free-text answers is hard

01

No single right answer

Chatbots reply in open text. Two very different answers can both be correct.

02

BLEU and ROUGE miss meaning

They count shared words. They can’t tell if an answer is useful, true or polite.

03

Humans don’t scale

Human review is the gold standard, but slow and costly for thousands of answers.

One LLM grades another on three criteria

Criterion (1–5)	Question the judge asks	A low-scoring answer
Helpfulness	Does it solve the user's need?	"Use a loop or something."
Correctness	Are the facts and steps true?	"The capital of Australia is Sydney."
Tone	Is it polite and suitable?	"Seriously? Look it up yourself."

Then we **validate** the judge: do its scores match what humans gave the same answers?

The pipeline in four steps

STEP 1

Data

Questions, answers and human scores

`prepare_data.py`

STEP 2

LLM judge

Rubric prompt gives 1–5 scores + a reason

`judge.py`

STEP 3

Compare

Spearman, Pearson, kappa, MAE vs humans

`evaluate.py`

STEP 4

Report

Tables, charts and the PDF report

`make_docs.py`

Plain Python: pandas, SciPy, scikit-learn, Matplotlib. The judge is called through Groq’s API.

Two datasets with real human ratings

120 HelpSteer2 answers

- Public NVIDIA dataset (2024)
- Rated 0–4 by trained annotators for helpfulness and correctness
- Short, single-turn samples, 24 from each score level

30 Our own answers

- 10 questions × 3 answers we wrote
- Good, polite-but-wrong, correct-but-rude, vague
- Rated by our team, needed because HelpSteer2 has no tone label

The judge: a model plus a rubric

```
You are a strict but fair evaluator.
Rate 1 (worst) to 5 (best):
HELPFULNESS 1 useless … 5 complete
CORRECTNESS 1 wrong … 5 fully correct
TONE 1 rude … 5 polite
Judge each one separately.
Do not reward length.
Reply only with JSON:
{"helpfulness": 4, "correctness": 5,
"tone": 3, "reason": "..."}
```

gpt-oss-120b

OpenAI's open-weight model, free through Groq, low reasoning effort

Temperature 0

Same input, same scores: results can be repeated

Robust parsing

Regex finds the JSON, scores are clipped to 1–5, failed calls retry

The judge agrees well with humans

0.69

Helpfulness, HelpSteer2 (ρ)

0.67

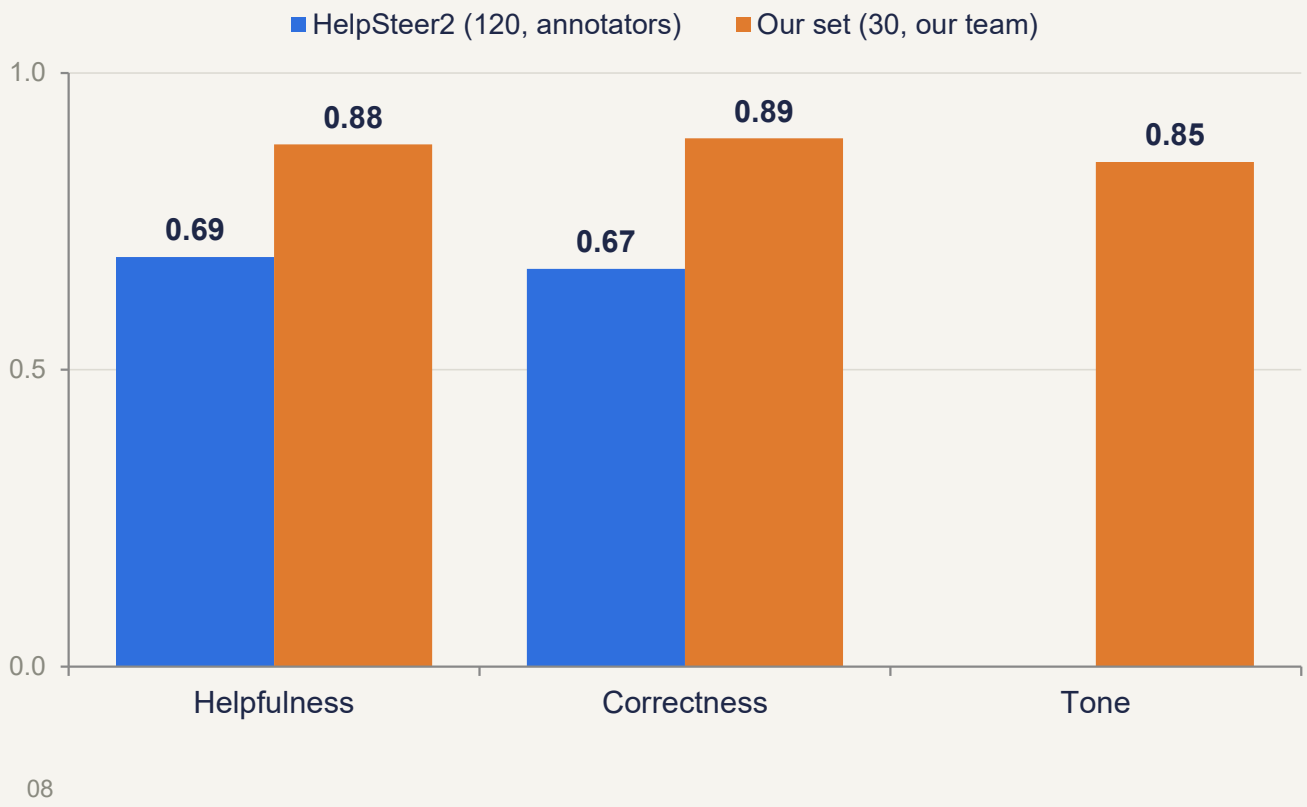
Correctness, HelpSteer2 (ρ)

0.85

Tone, our set (ρ)

Dataset	Criterion	Spearman ρ	Kappa	Within 1 pt	Bias
HelpSteer2	Helpfulness	0.69	0.69	84%	+0.01
HelpSteer2	Correctness	0.67	0.67	79%	+0.08
Our set	Helpfulness	0.88	0.87	93%	+0.23
Our set	Correctness	0.89	0.85	90%	−0.03
Our set	Tone	0.85	0.88	93%	+0.03

Strong agreement on every criterion



Spearman ρ between judge and human scores. Higher on our set, whose answers are clearly good or clearly bad.

No tone labels in HelpSteer2, so tone is only measured on our set.

Where the judge goes wrong

TOO GENEROUS

“How do I make a Trident cocktail?”

A fluent, confident recipe. The judge called it “clear, accurate”; the annotators rated it wrong.

Judge 5 / 5 **Humans** 1 / 1

TOO STRICT

“Tag each new concept with [concepts]…”

The judge said the answer ignored the request; the annotators found it helpful.

Judge 1 / 2 **Humans** 4 / 5

Of the 5 biggest disagreements, 2 were too generous and 3 too strict: **no one-sided bias**, but the judge cannot fact-check what it doesn’t know.

Takeaways

What we showed

- An LLM judge is fast, cheap and repeatable
- $\rho \approx 0.7$ with real annotators, near-zero bias
- It can score tone separately from facts

Limits and next steps

- Short single-turn samples, one judge model
- Tone checked on 30 answers by our team
- Next: more data, more judges, more raters

Thank you. Questions?